

2017-Ph-H1-Q18

Section: Particles and Waves

Topic: Inverse Square Law

Detector is moved further from a radiation source. What happens to count rate?

 **Inverse Square Law:**

$$I \propto \frac{1}{r^2}$$

If distance **doubles**, intensity (and count rate) drops to $\frac{1}{4}$

 **Final Answer: A — $\frac{1}{4}$**

 **Revision Tips:**

- Applies to point sources spreading energy (light, radiation, sound)
- Double distance = $\frac{1}{4}$ intensity